1 7/64" with four-valve heads, others measure .703" (approx. 45/64"). Refer to figure 9.

LINER MAINTENANCE

Cylinder liners will render satisfactory service for extended periods if the engine has proper care. Wear on a cylinder liner and piston is directly related to the amount of abrasive dust and dirt introduced into the engine combustion chambers via air intake. Dust so introduced and combined with lubricating oil on the cylinder walls forms an ideal lapping compound. To avoid such a condition, the vehicle air cleaners should be serviced regularly as instructed in the maintenance manual covering the specific vehicle.

When the clearance between the piston and cylinder liner becomes excessive the performance of the engine will be unsatisfactory. Replacement or reconditioning of liners and associated parts will be necessary.

When a cylinder liner change is necessary, due to either wear or scoring, the old liner may be removed from the cylinder block and a new liner installed, or the old liner may be honed oversize and reinstalled. New cylinder liners are available only in the standard size inside diameter; however, .005", .010", .020", and .030" oversize outside diameter liners are available for use in cylinder block bores which have been oversized to correct distortion or out-of-roundness.

CLEANING LINER PORTS

Whenever 1000 hour inspection indicates that air ports in liners require cleaning, the operation can be performed as follows:

1. Remove hand hole covers at side of cylinder block, also remove blower assembly.

2. Use suitable tool to remove deposits from liner ports, being careful that liner is not damaged.

3. Use vacuum to lift all carbon deposits from air box, then use compressed air to remove all remaining particles.

4. Inspect air box drains to be sure they are clean.

5. Install blower assembly and hand hole covers as directed in respective sections of this manual.

6. Whenever cylinder liners have been removed, the air ports can be cleaned by soaking in a hot caustic soda or lye solution to loosen carbon deposits. Final cleaning can then be accomplished by brushing loose deposits.

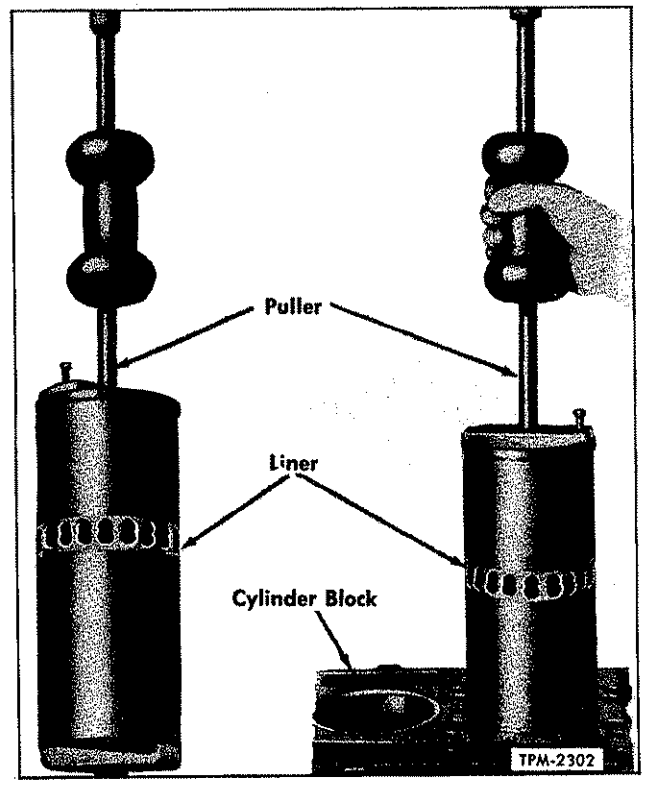


Figure 10—Removing Cylinder Liner

LINER REMOVAL

To remove a cylinder liner, the following preliminary operations will be necessary.

1. Remove the cylinder head assembly as instructed in CYLINDER HEAD section of this manual.

2. Remove the oil pan.

3. If necessary, remove the lubricating oil pump and discharge line assembly together with the pump drive shaft. Refer to LUBRICATING OIL PUMP in LUBRICATION section of this manual.

4. Remove cylinder liner compression gasket from each liner.

5. Scrape carbon from the upper inner surface of cylinder liner, before attempting to remove piston assembly.

6. Remove piston and connecting rod assembly. Refer to PISTON AND CONNECTING ROD section later in this manual.

7. Since the cylinder liners are a loose fit, they may be removed easily from the top of the cylinder block by hand, after loosening with the tool illustrated in figure 10. Use tool in following manner:

(a) Slip lower puller clamp upward on puller rod and off tapered cone. Cock lower clamp on